

Appendix A - Geographe Marine Park Natural Values Report

South-west Marine Parks Network

Table of contents

Natural Values	1
Sampling summary	2
Benthic ecosystem extent	3
Observations	3
Dominant habitat types	4
Habitat distribution and probability	5
Functional reef distribution and probability	6
Benchmarks of benthic natural values	7
Seagrass	7
Macroalgae	8
Sessile invertebrates	9
Benchmarks of demersal fish natural values	10
Survey effort	10
Species richness	11
Total abundance	13
Large Reef Fish Index*	15
Reef fish thermal index	17

Natural Values

Understanding of natural values

Sampling summary

Table 1: Summary of monitoring campaigns in the study area, showing the survey dates, sampling method, management zones sampled, and the number of samples with each data type. NPZ = National Park Zone, HPZ = Habitat Protection Zone, MUZ = Multiple Use Zone, SPZ = Special Purpose Zone, SZ = Sanctuary Zone, GUZ = General Use Zone.

Date	Campaign name	Method	Areas	Samples	Habitat	Fish	
					Count	Count	Length
Dec 2014	2014-12 Geographe-bay stereo-BRUVs	BRUV	Coastal waters, GUZ, HPZ, MUZ, NPZ, SPZ, SZ	324	320	297	238
Mar-Apr 2024	2024-04 Geographe stereo-BRUVs	BRUV	Coastal waters, GUZ, HPZ, MUZ, NPZ, SPZ, SZ	261	260	259	246

Benthic ecosystem extent

Observations

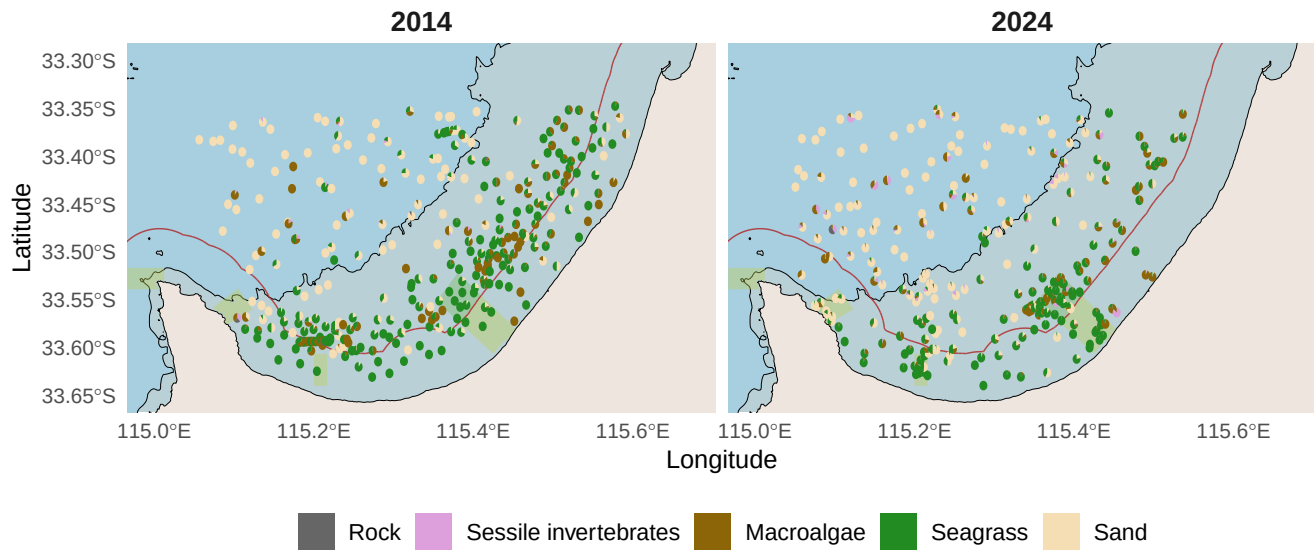


Figure 1: Habitat classes observed in stereo-BRUV imagery. Habitat classes observed in spatially balanced stereo-BRUV and drop-camera imagery visualised as spatial pie charts. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Dominant habitat types

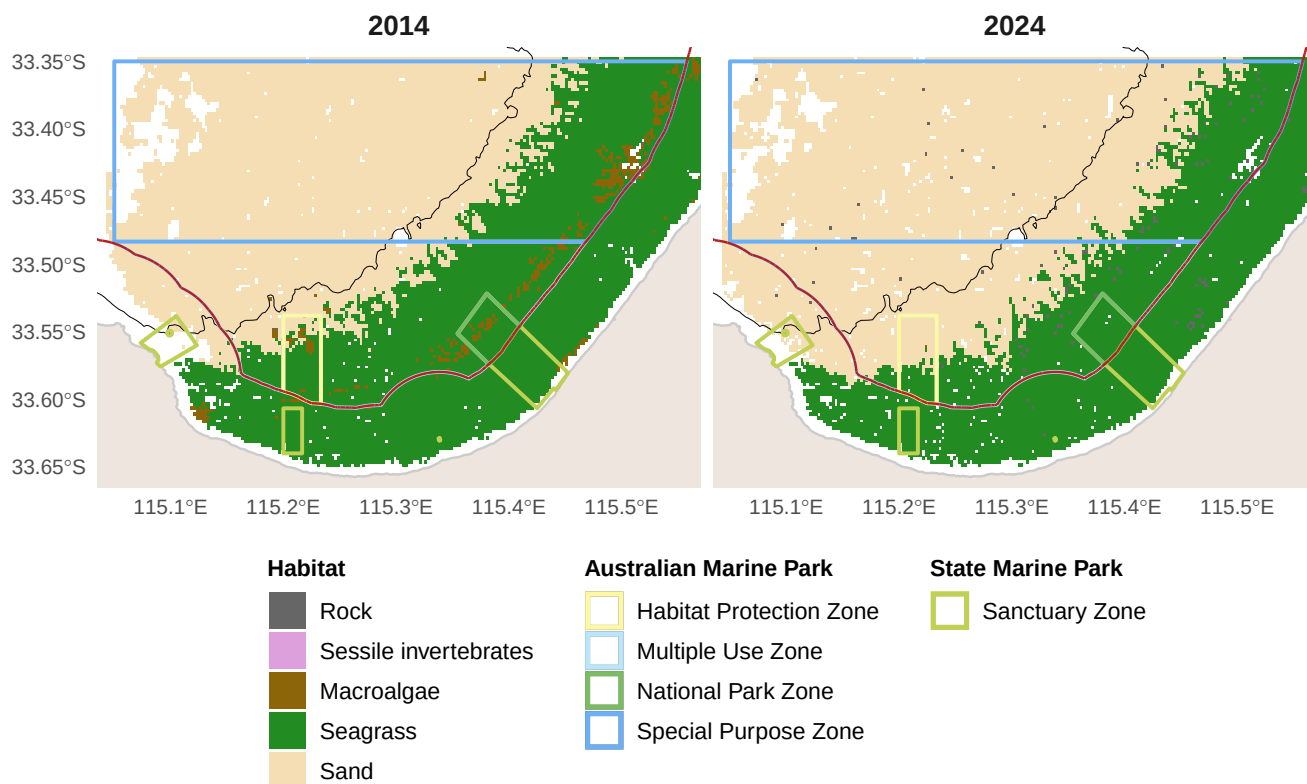


Figure 2: Predicted dominant habitat type from stereo-BRUV groundtruthing. The green polygon represents the extent of no-take National Park Zones and State no-take sanctuary zones. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Habitat distribution and probability

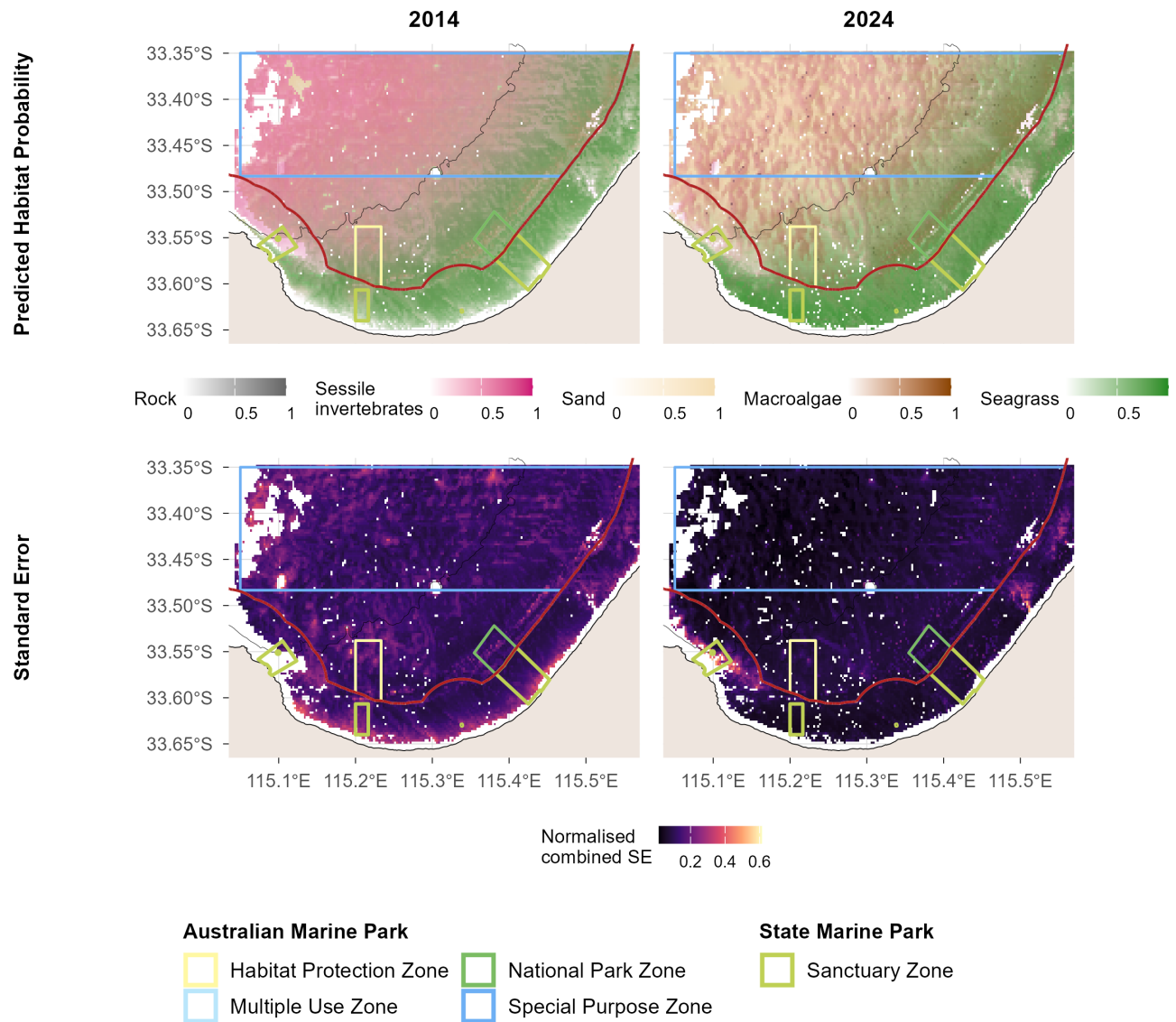


Figure 3: Predicted habitat distribution probability and uncertainty from stereo-BRUV groundtruthing. The green polygon represents the extent of no-take National Park Zones. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Functional reef distribution and probability

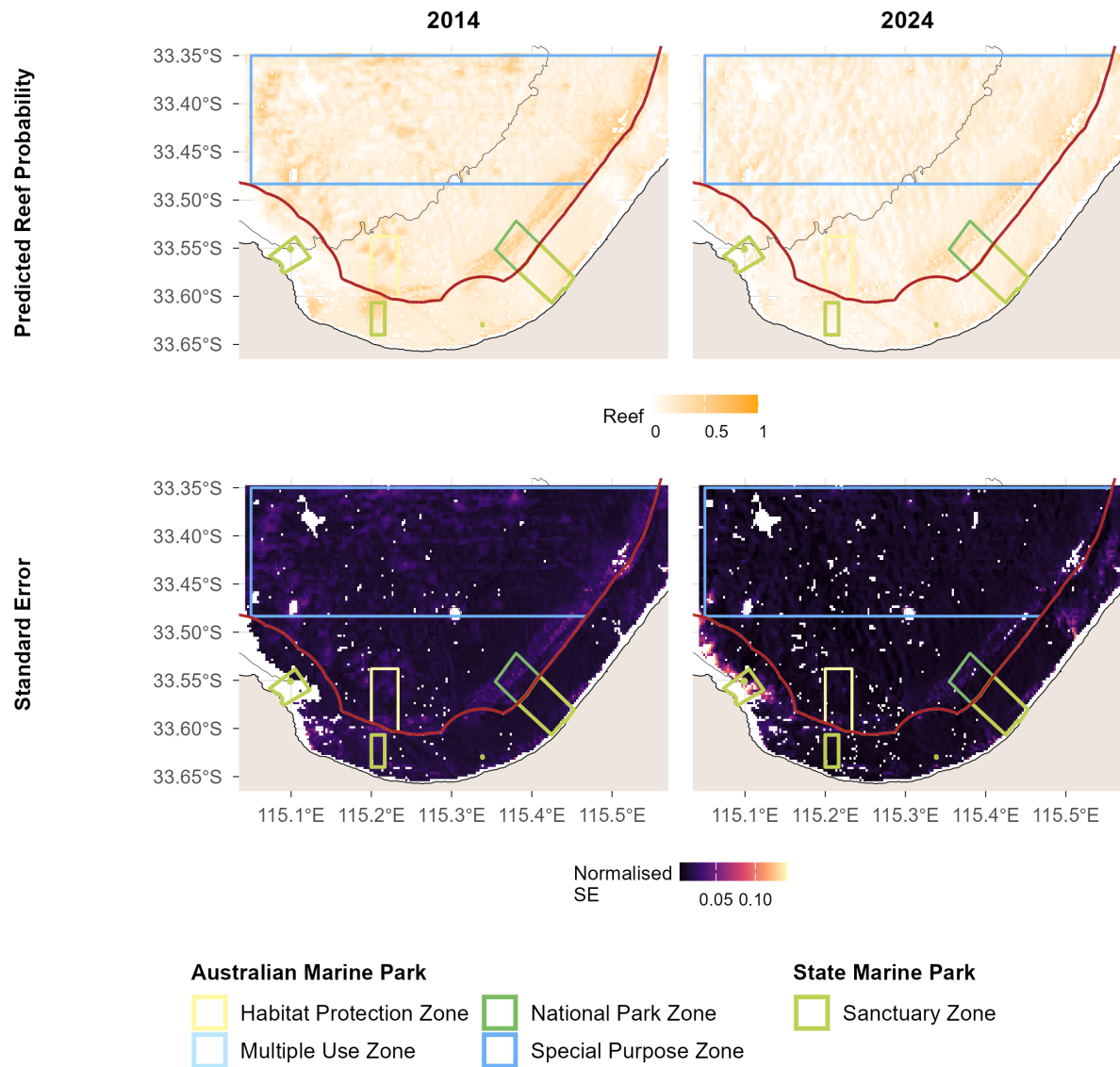


Figure 4: Predicted reef distribution probability and uncertainty from stereo-BRUV groundtruthing. The green polygon represents the extent of no-take National Park Zones. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Benchmarks of benthic natural values

Seagrass

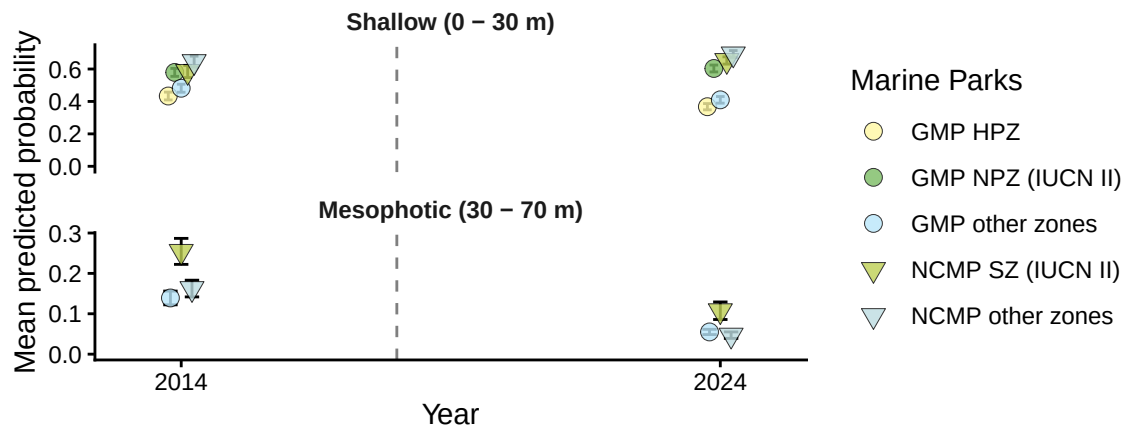


Figure 5.1: Seagrass: temporal plots by depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

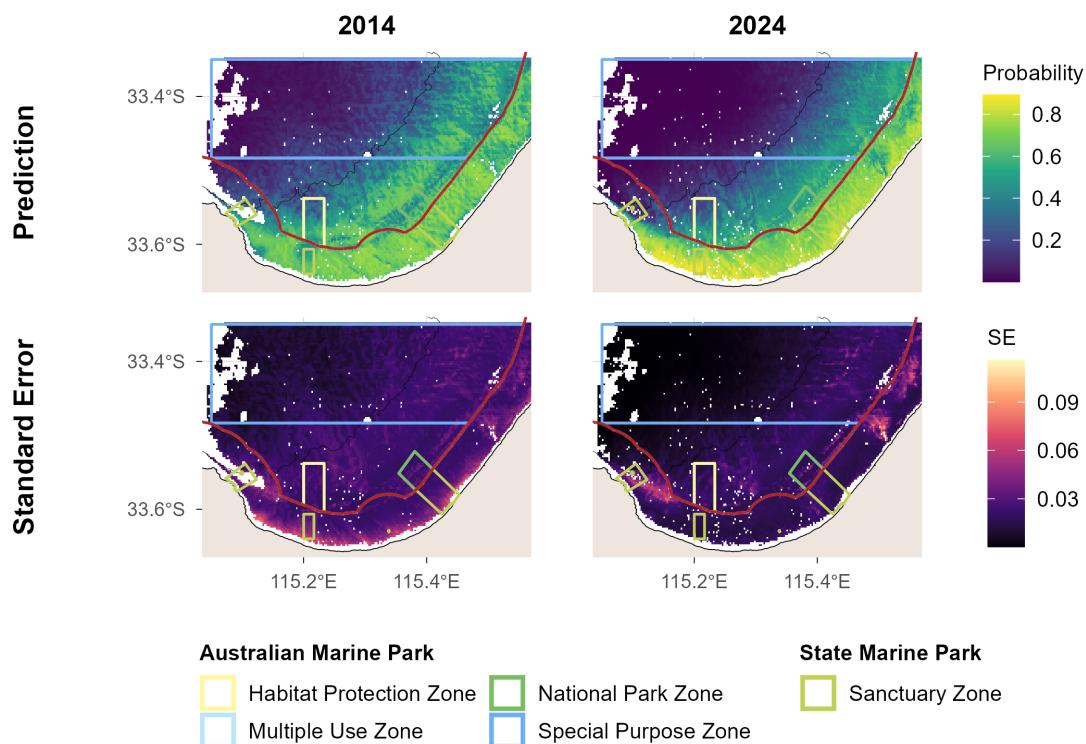


Figure 5.2: Seagrass: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Macroalgae

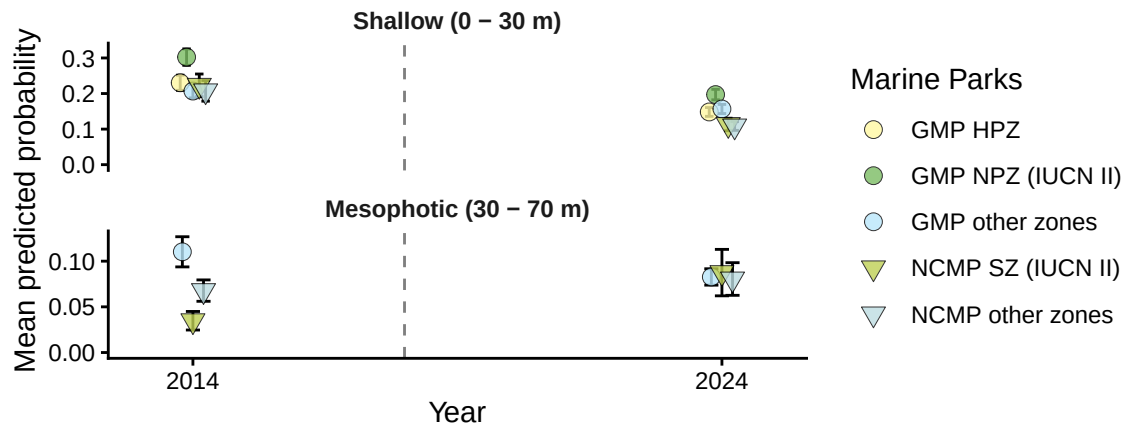


Figure 6.1: Macroalgae: temporal plots by depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

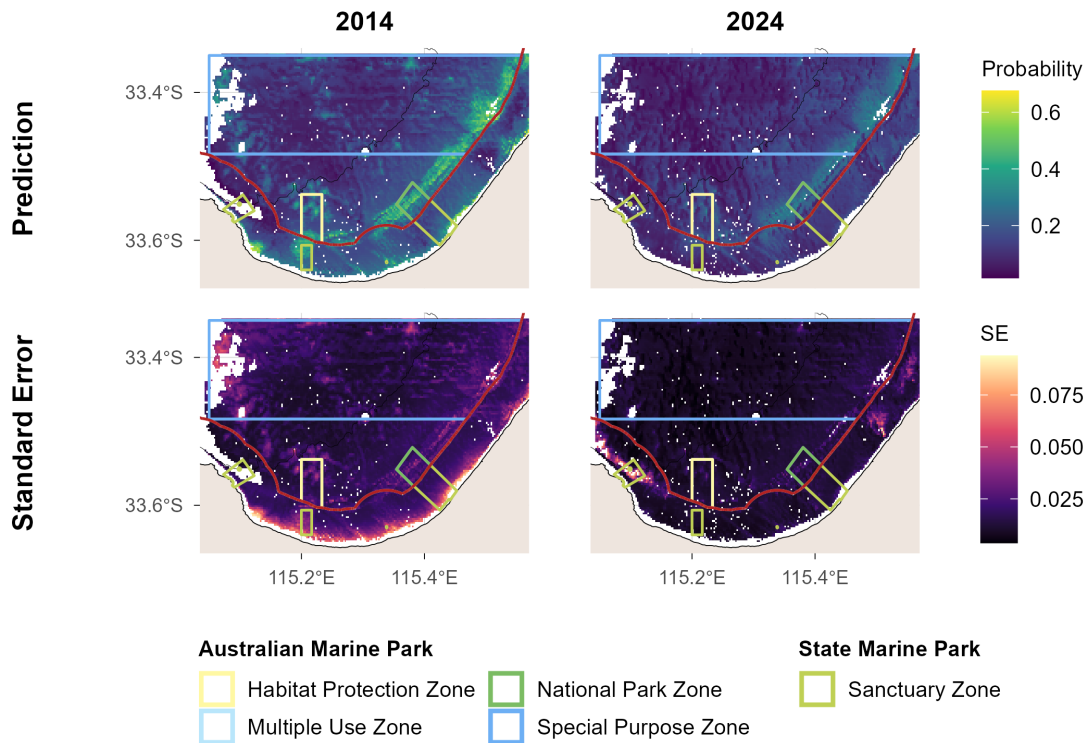


Figure 6.2: Macroalgae: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Sessile invertebrates

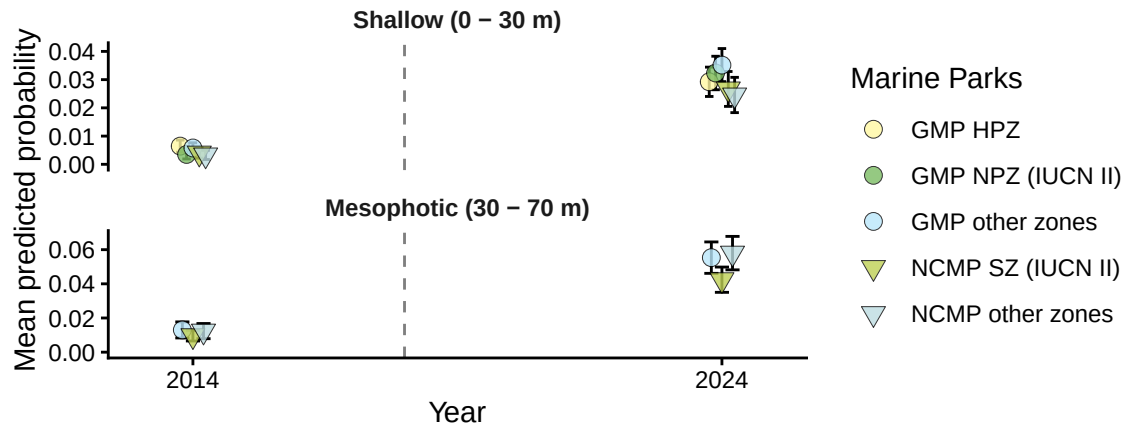


Figure 7.1: Sessile invertebrates: temporal plots by depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

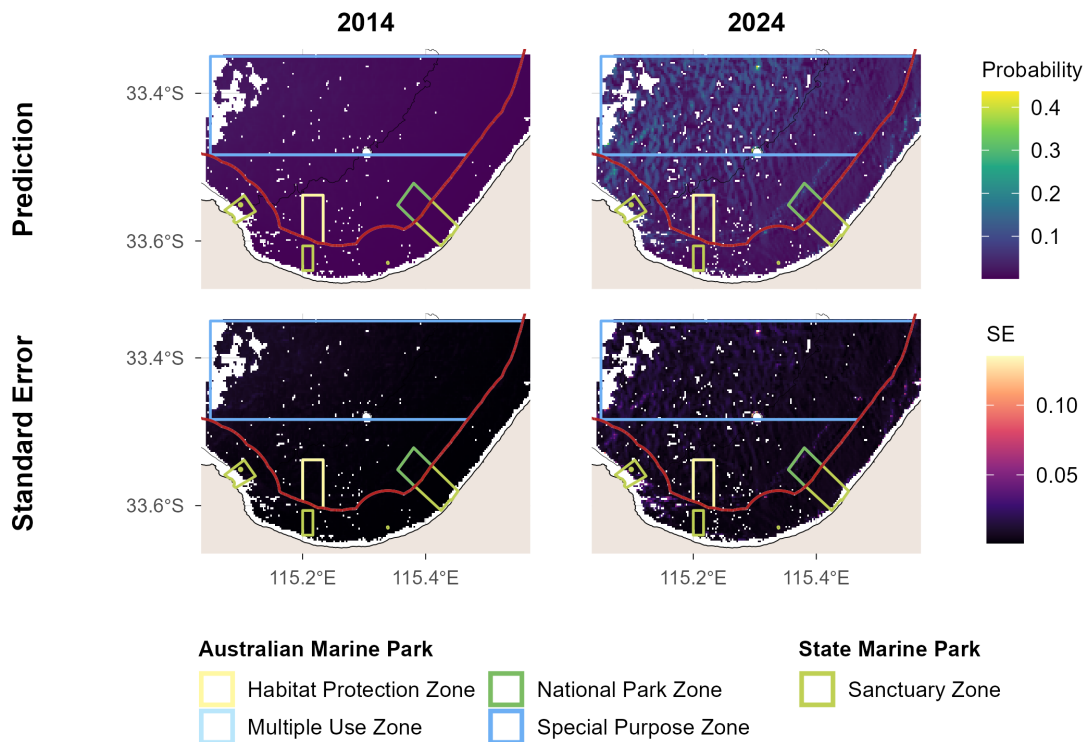
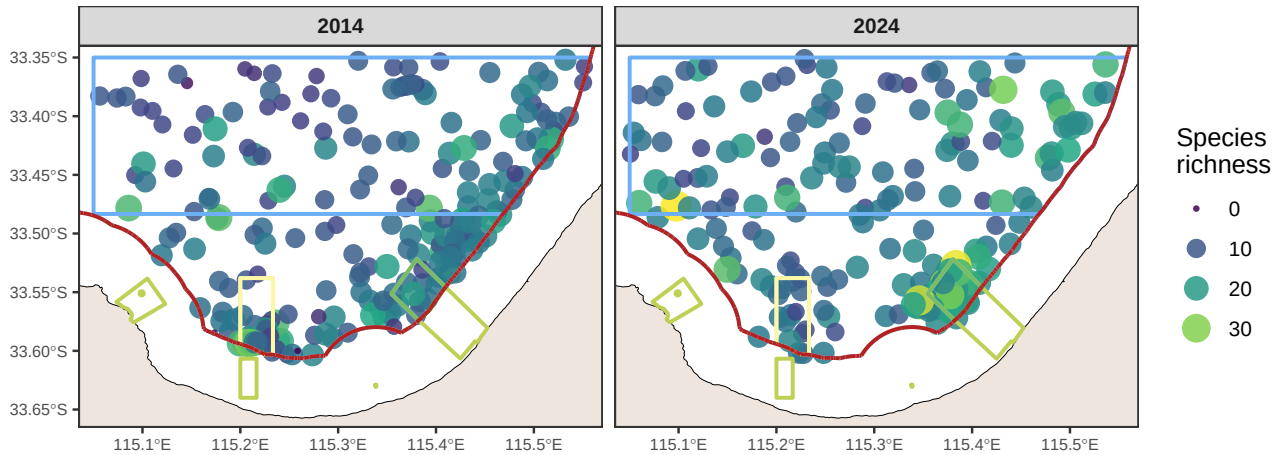


Figure 7.2: Sessile invertebrates: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

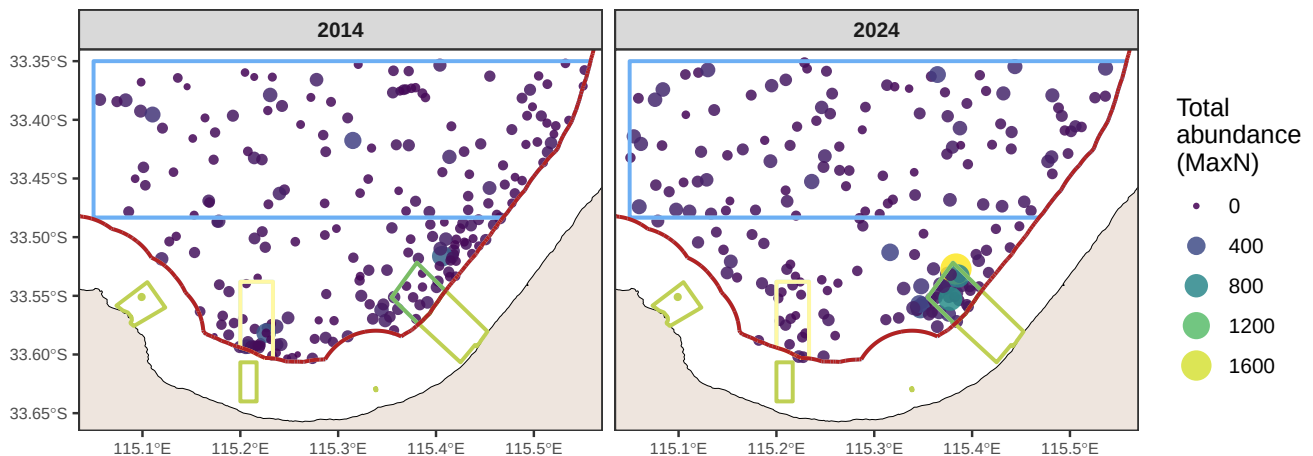
Benchmarks of demersal fish natural values

Survey effort

a)



b)



Australian Marine Park

- Habitat Protection Zone
- Multiple Use Zone

- National Park Zone
- Special Purpose Zone

State Marine Park

- Sanctuary Zone

Figure 8: Spatial distribution of observed species richness and total abundance (MaxN) per stereo-BRUV drop, pooled across all deployments. State and Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters.

Species richness

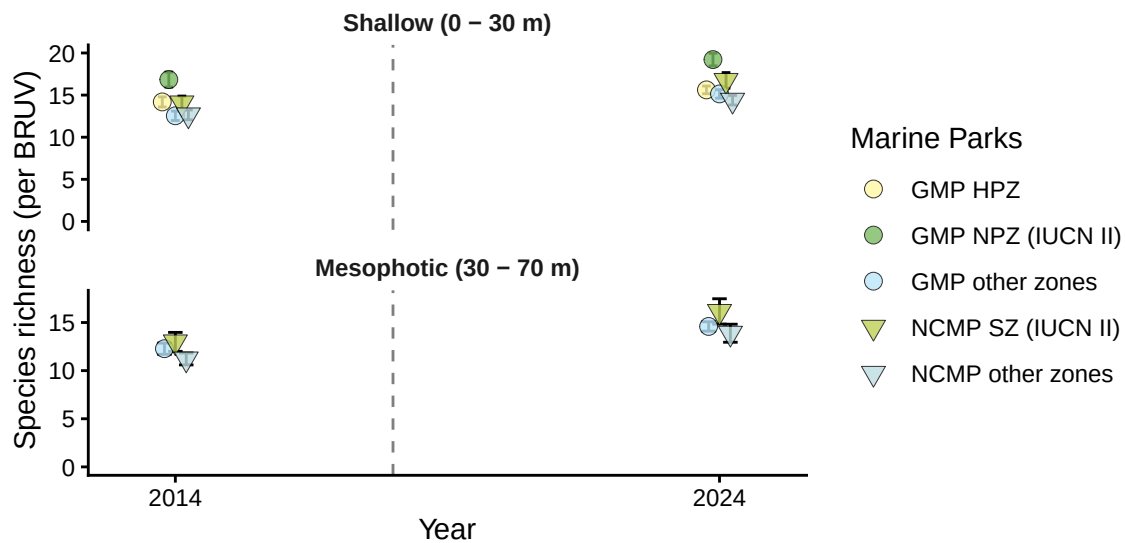


Figure 9.1: Demersal fish - Species richness: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

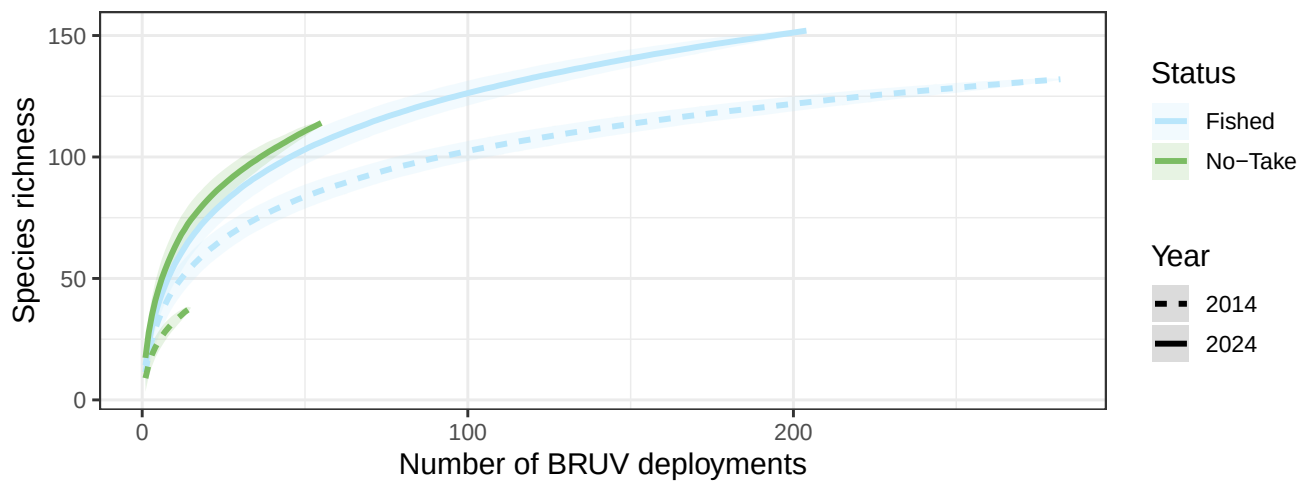


Figure 9.2: Demersal fish - Species richness: species accumulation curve by zone type from stereo-BRUV groundtruthing.

Species richness

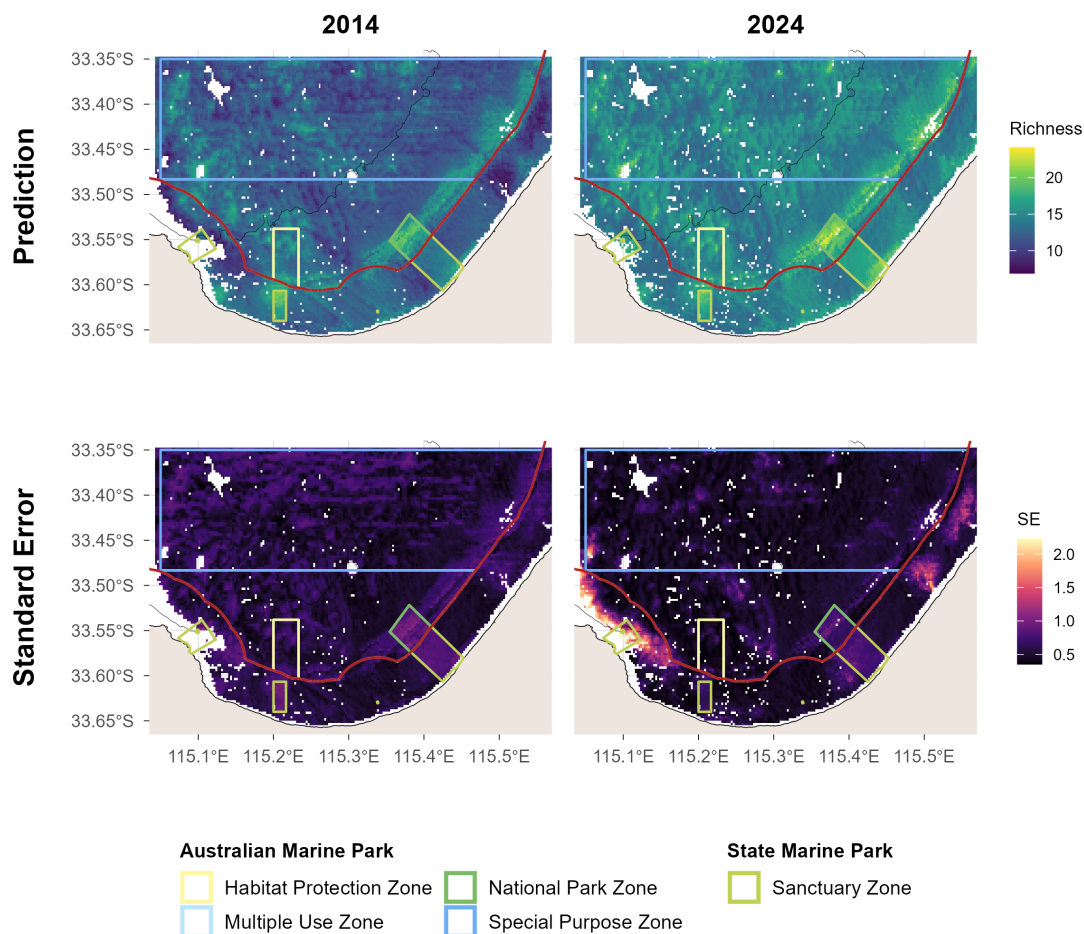


Figure 9.3: Demersal fish - Species richness: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Total abundance

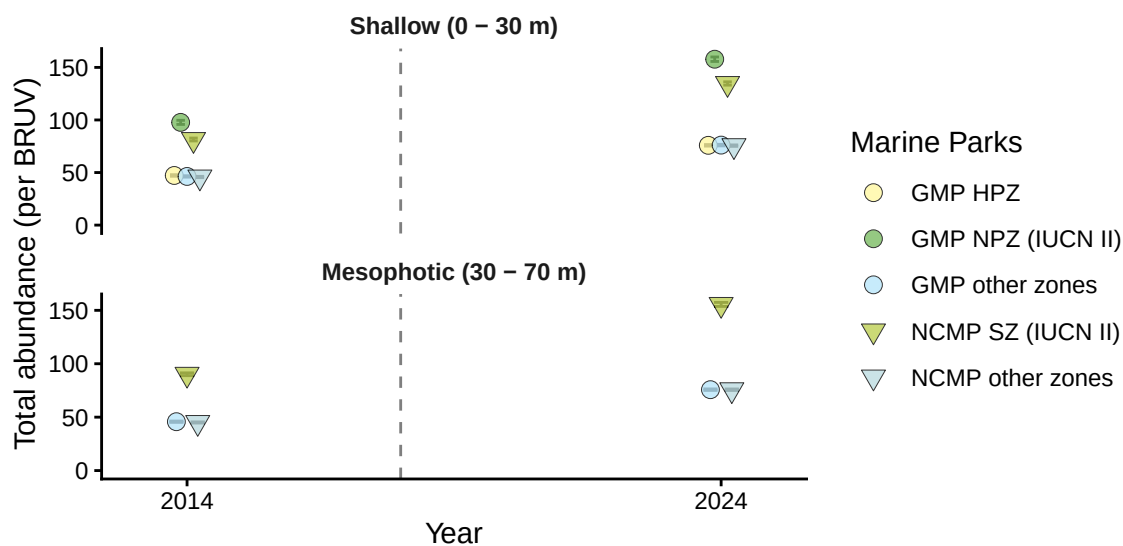


Figure 10.1: Demersal fish - Total abundance: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

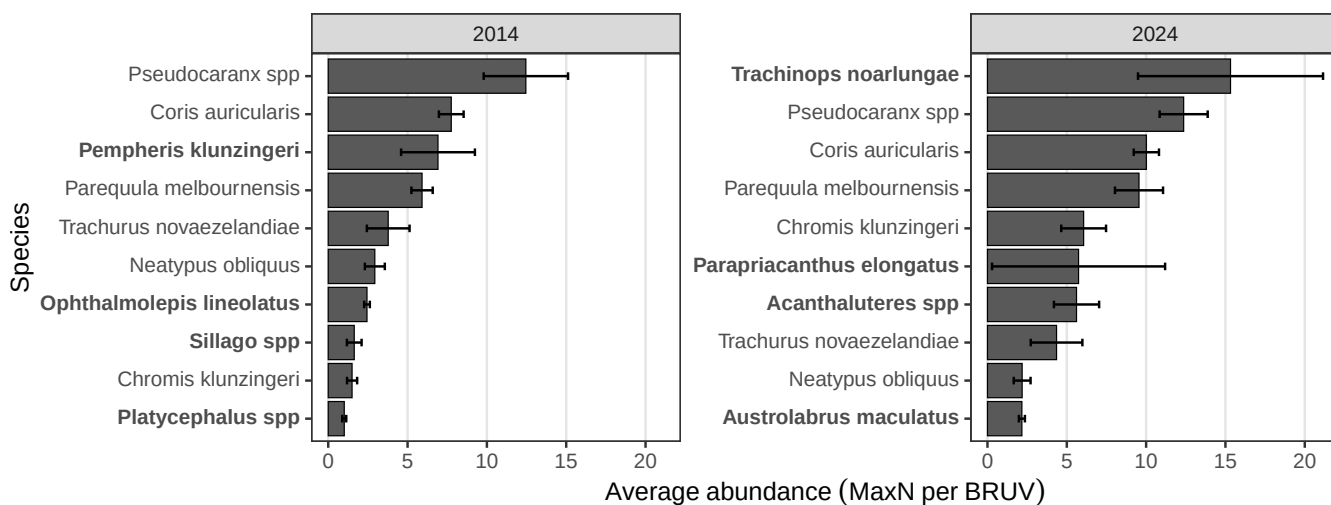


Figure 10.2: Demersal fish - Total abundance: top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. **Bold** species are unique within the 10 most abundant species by year.

Total abundance

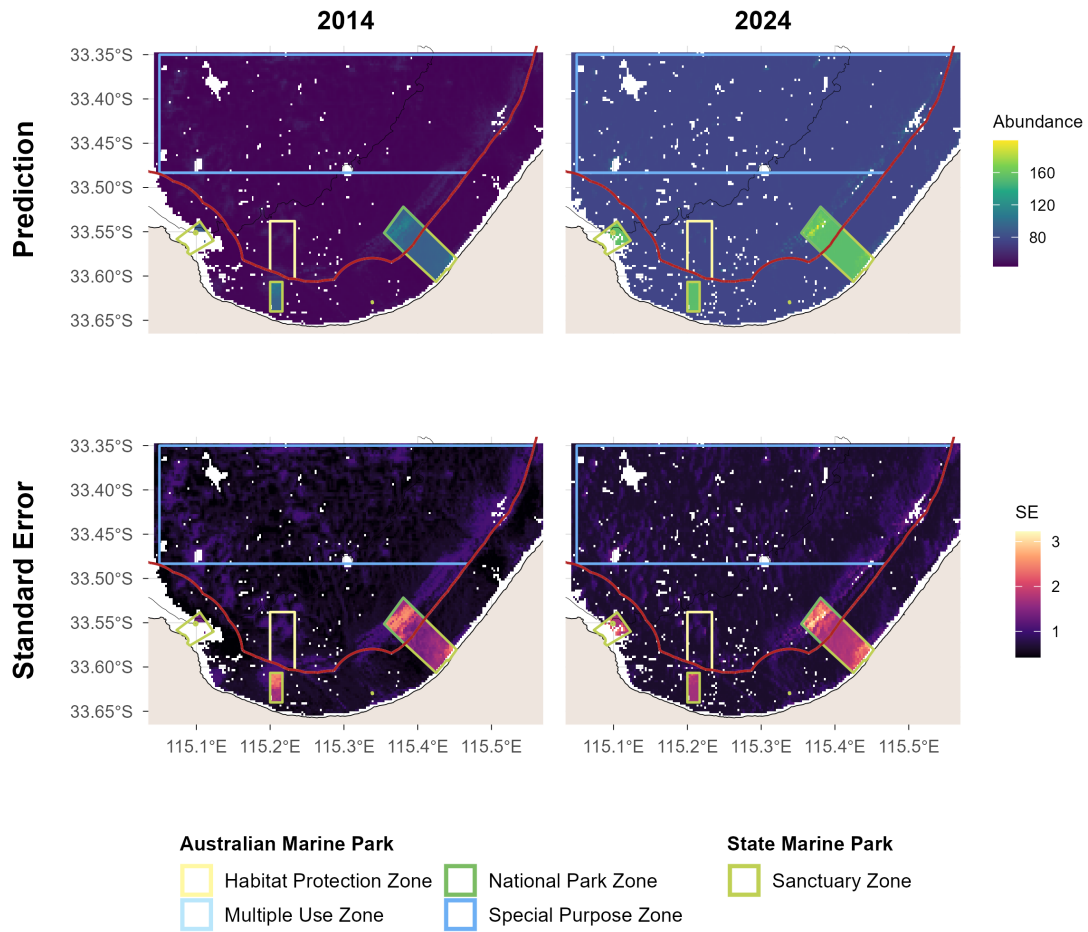


Figure 10.3: Demersal fish - Total abundance: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Large Reef Fish Index*

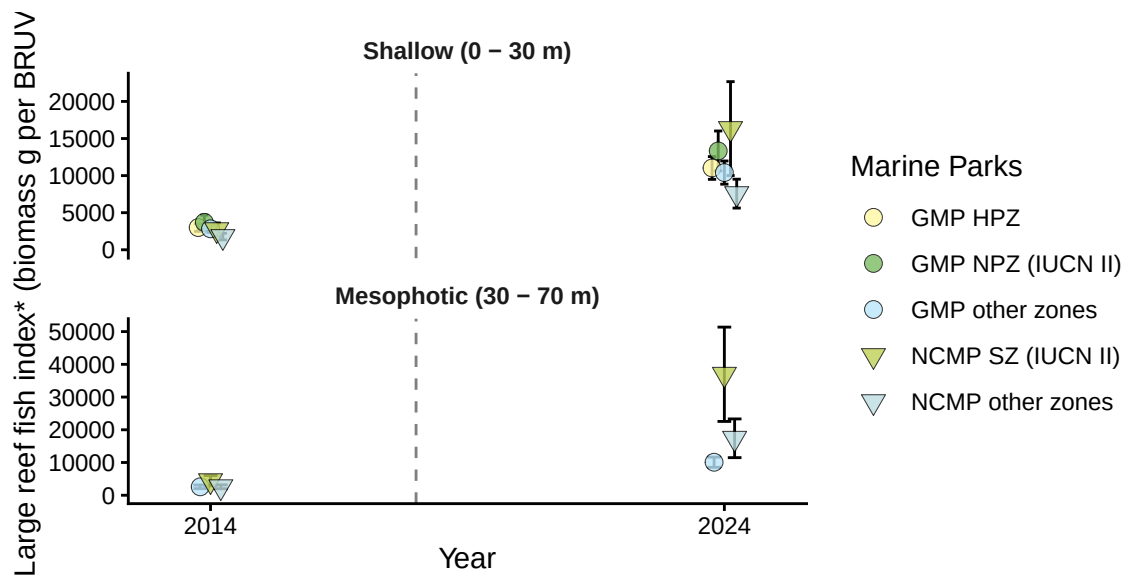


Figure 11.1: Demersal fish - Large Reef Fish Index*: benchmark plots by zone and depth contour from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone. *Large Reef Fish Index representing the biomass of all fish greater than 20 cm (B20) has been adapted here for stereo-BRUV data by removing sharks and rays.

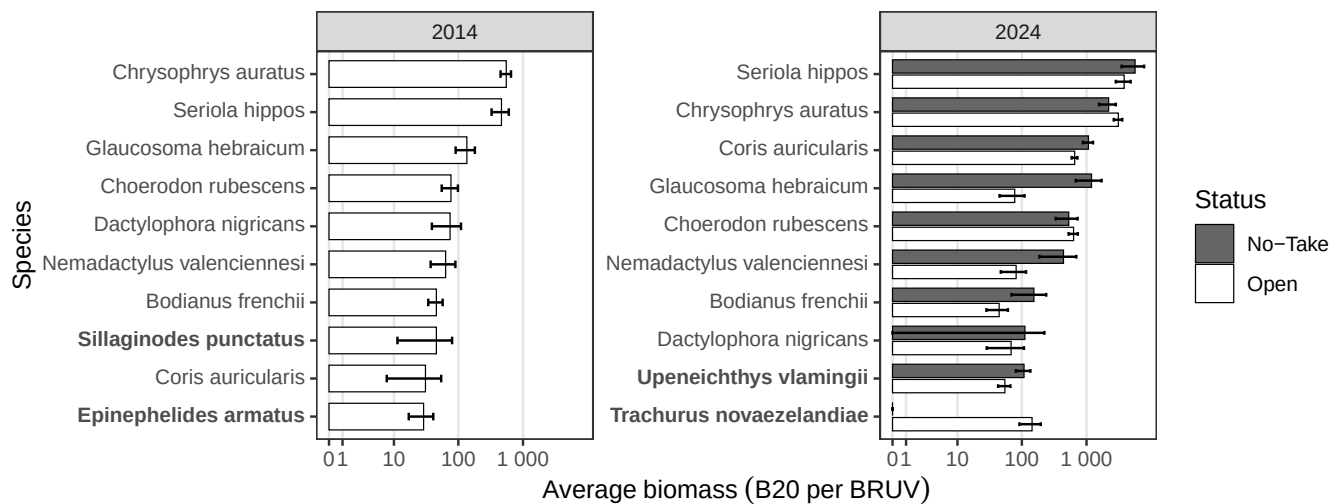


Figure 11.2: Demersal fish - Large Reef Fish Index*: top 10 greatest biomass species by status (pooled across deployments) from stereo-BRUV groundtruthing. **Bold** species are unique within the 10 greatest biomass species by year.

Large Reef Fish Index*

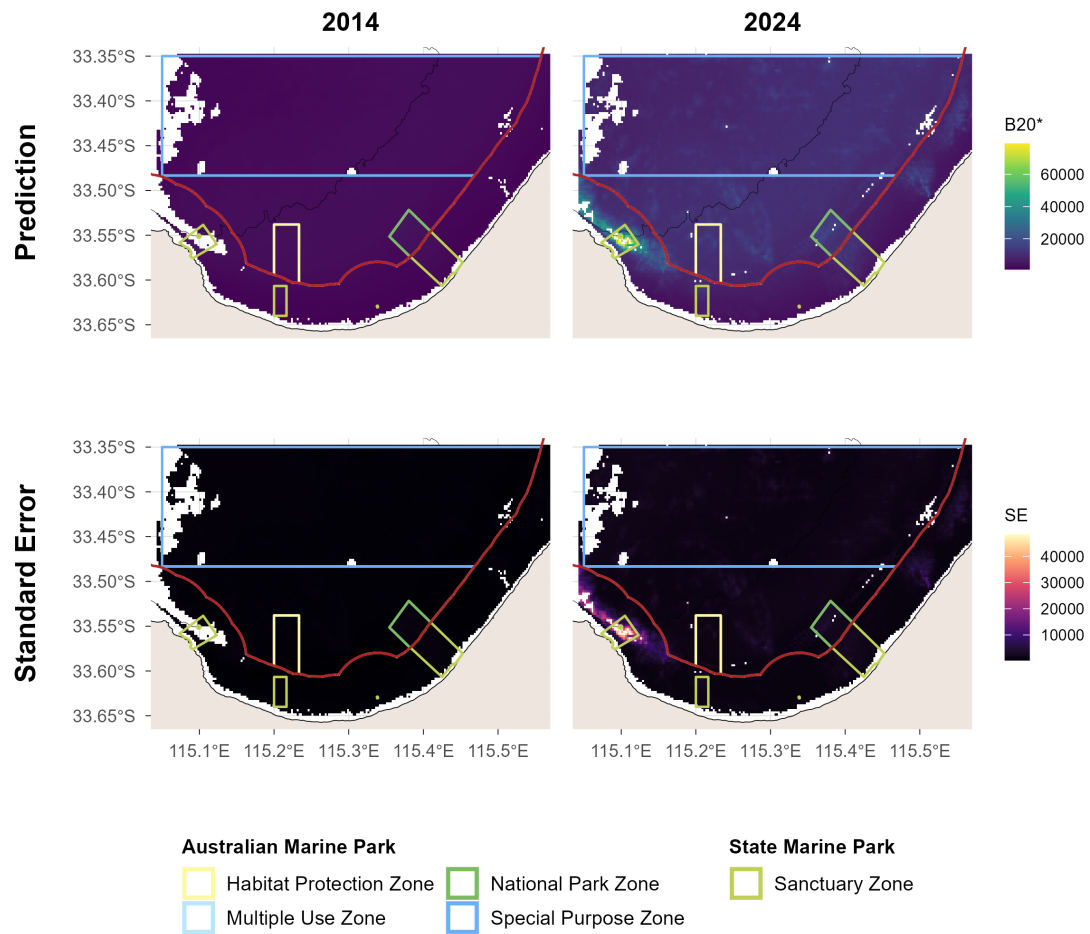


Figure 11.3: Demersal fish - Large Reef Fish Index*: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.

Reef fish thermal index

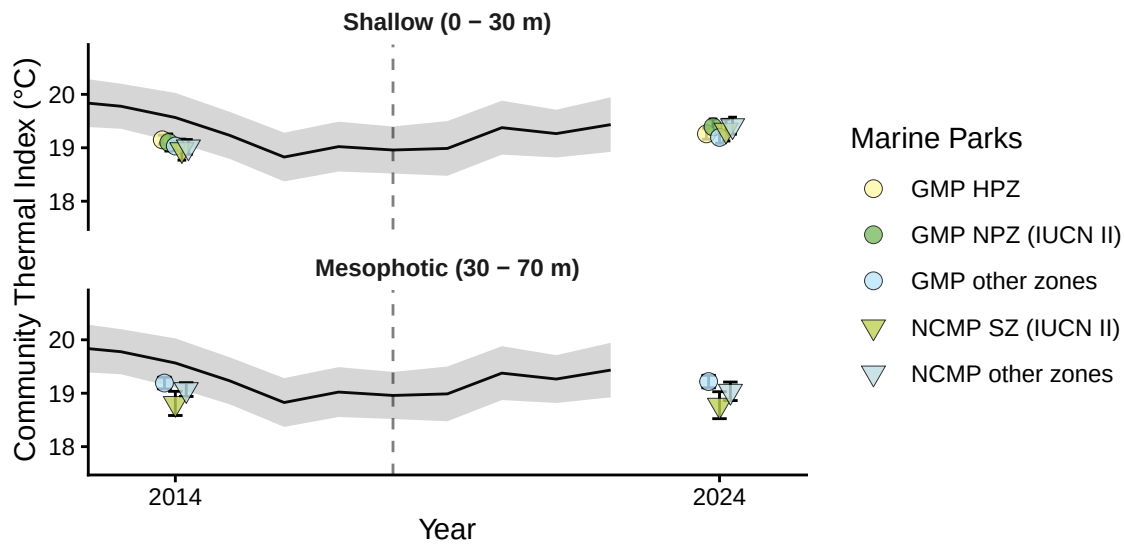


Figure 12.1: Demersal fish - Reef Fish Thermal Index: benchmark plots by zone and depth contour, with monthly sea surface temperature (black line, grey band = SD) overlaid, from stereo-BRUV groundtruthing. Dashed vertical line indicates the date of establishment of the marine park. GMP = GMP (Commonwealth), HPZ = Habitat Protection Zone, NPZ = National Park Zone, NCMP = Ngari Capes Marine Park (State), SZ = Sanctuary Zone.

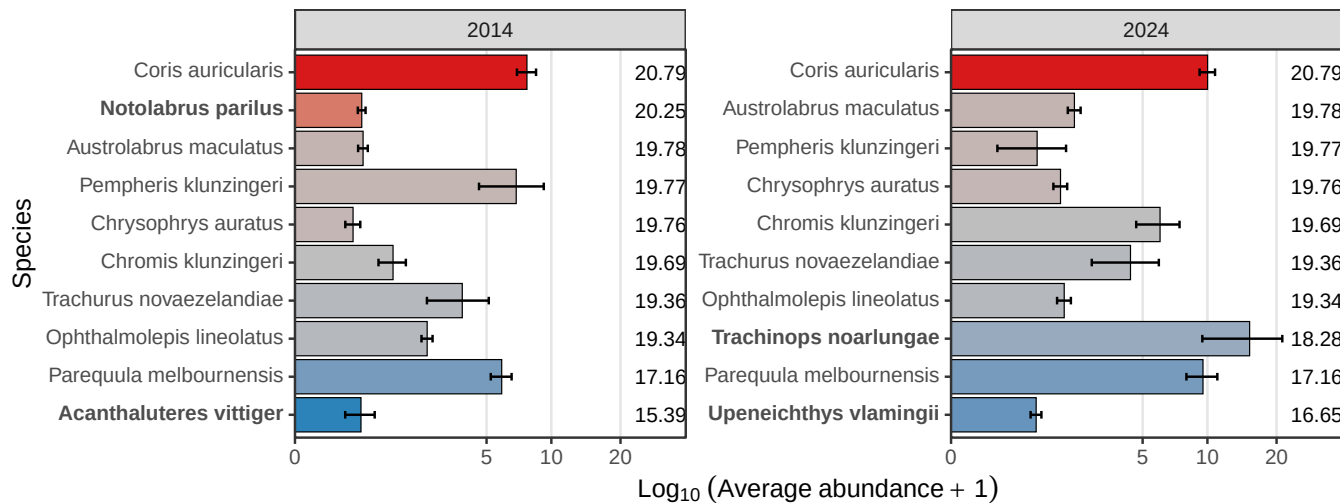


Figure 12.2: Demersal fish - Reef Fish Thermal Index: species thermal niche of top 10 most abundant species (pooled across deployments) from stereo-BRUV groundtruthing. **Bold** species are unique within the 10 most abundant species by year.

Reef fish thermal index

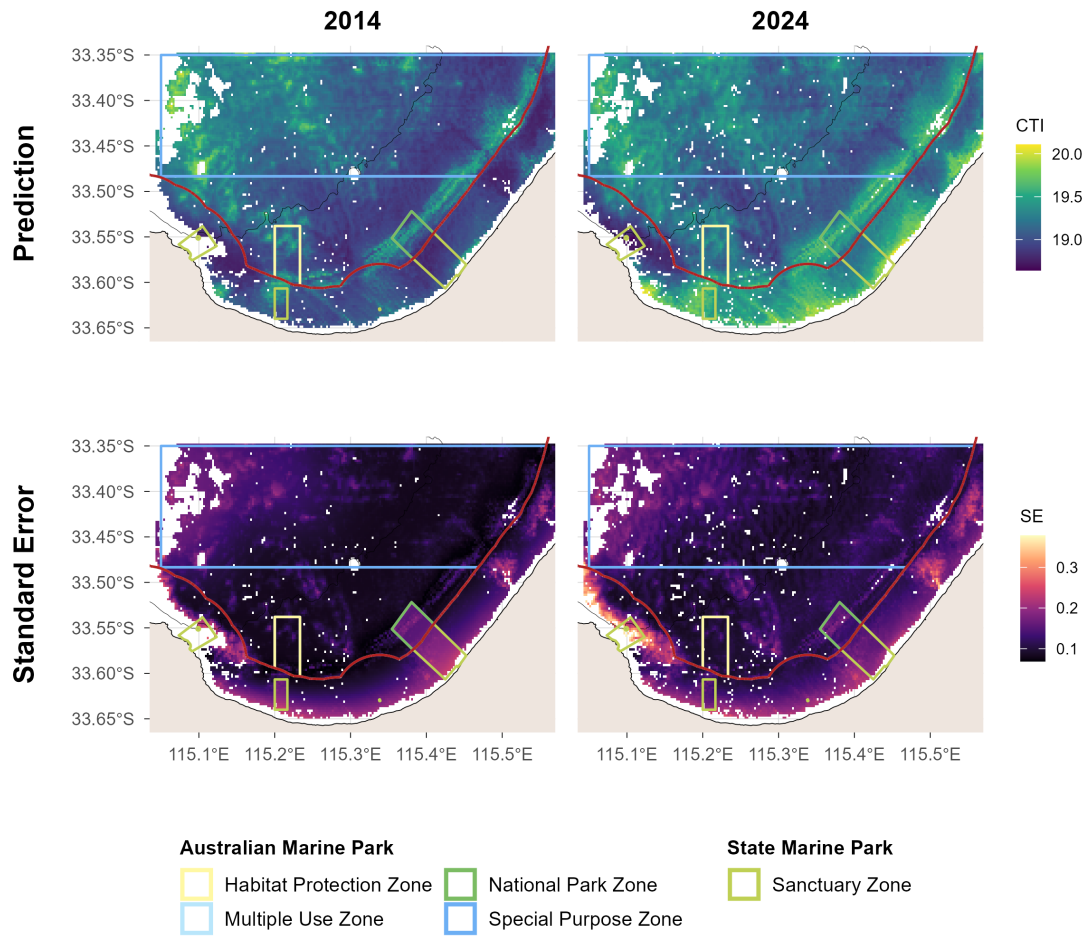


Figure 12.3: Demersal fish - Reef Fish Thermal Index: spatial plots of distribution probability and uncertainty from stereo-BRUV groundtruthing. Commonwealth marine park boundaries are shown. The red line delimits state and commonwealth waters. Bathymetric contour at 30 m is shown.